

Dr. Anthony Hauser
Unisanté, Center for Primary Care and Public Health
University of Lausanne
Rte de la Corniche 21
CH-1010 Lausanne, Switzerland

To the editors of the European Journal
of Epidemiology

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Dear editors,

We are pleased to submit our manuscript entitled “COVID-19 disrupted patterns of cause-specific mortality in Switzerland” for consideration in *the European Journal of Epidemiology*. Using high-quality national mortality data and a Bayesian multivariate Poisson model, we quantify cause-specific excess mortality during 2020–2021 and reveal how deaths from different causes are interdependent.

We analyzed weekly deaths in Switzerland from 2011–2021 by age, sex, region, and nine cause groups, accounting for seasonality, long-term trends, population changes, and cross-cause correlations. Cardiovascular deaths peaked during COVID-19 waves, while no cause showed sustained excess over the full period. In individuals ≥ 80 years, declines in non-COVID respiratory (-25%) and mental/neurological (-12%) deaths were largely offset by COVID-19 mortality, resulting in only a modest all-cause excess. We also identified strong pre-pandemic correlations between respiratory, cardiovascular and mental/neurological deaths ($\rho > 0.8$), and observed that excess mortality across these causes remained closely correlated during the pandemic.

Our framework formally models these cross-cause correlations and leverages the COVID-19 pandemic as a natural experiment to disentangle mechanisms that have historically overlapped. We identified three key drivers of cause-specific disruptions: misclassification of deaths, mortality displacement, and effects of control measures.

Our work builds on recent discussions in the European Journal of Epidemiology. Nilsson et al. quantified cardiovascular excess and deficits in respiratory and dementia deaths in Nordic countries (1). Li et al. emphasized the importance of accounting for co-dependence between causes when assessing indirect effects of COVID-19 on mortality (2). Jasilionis et al. examined the role of cardiovascular disease in overall mortality and prompted discussion on using cause-specific deaths to explore mortality redistribution (3–5). By formally modelling cross-cause correlations, our manuscript provides both methodological innovation and empirical insight directly relevant to the journal’s scope.

Thank you for your time and consideration. We look forward to your response.

Sincerely,

Dr. Anthony Hauser on behalf of all authors



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